Byung-cheol Kang

Genomics · Bacteriophages · Research Pipelines

A bioinformatics researcher and developer connecting biological questions with analysis design and interpretation. My work extends from microbes to humans, integrating diverse analytical approaches and building research pipelines and databases. AI-powered coding accelerates my research and amplifies my ability to automate workflows and develop analytical tools.

CURRENT AND RECENT ROLES

MicrobiotiX · Bioinformatician — 2026.07 - PRESENT

- Develop bacterial genome analysis workflows covering quality control, annotation, typing, antimicrobial resistance, and prophage detection to support production-strain assessment.
- Integrate phage genome quality, termini and packaging, taxonomy, lifestyle predictions, and tail-protein evidence into reports with traceable conclusions.
- Develop and extend analysis libraries and command-line tools combining experimental host-range matrices with sequence and protein features for clustering comparisons and cocktail candidate selection.
- Build and maintain a Klebsiella pneumoniae tail/RBP reference database linking sequences to K-type and KL-type evidence, with source citations and separate experimental and predicted labels.
- Develop in silico design tools that evaluate tail-protein candidates for specified capsule targets and generate candidate design sequences and review reports.
- Compare protein sequences, domains, predicted structures, and reference structures to support RBP annotation and assess the structural plausibility of replacement and fusion candidates.

Wittgenbio · Computational Biology Research — 2026.07 - 2026.10

- Built a multi-study CRC scRNA-seq atlas through metadata standardization, quality control, integration, clustering, and cell-type annotation.
- Cross-checked original author annotations, marker evidence, and tumor/normal metadata while tracking label decisions and source provenance across versions.
- Annotated externally supplied normal-tissue atlas data and combined tissue-specific expression with tumor and patient/specimen evidence to evaluate therapeutic-target candidates at the RNA level.

Past platform evidence. The integrated workflow and operational results in this portfolio describe work performed at Sanigen (2020.03 - 2025.11). They are not presented as MicrobiotiX or Wittgenbio assets or current-company outcomes.

PROJECT 01 · PLATFORM

Integrated Microbial Genomics Platform

PROBLEM

Separate assay scripts, environments, quality checks, server paths, and reporting conventions made analysis difficult to reproduce and review consistently.

SYSTEM DESIGN

Entry point	One Snakemake dispatcher selects 16S, ITS, WGS, or Hybrid profiles.
Execution	Server-specific YAML configuration with Conda, Docker, or Apptainer and pinned BioContainers.
Quality	Input validation, raw-read QC, assembly/analysis checks, review gates, and retained logs.
Delivery	Custom Python/R processing, visualizations, HTML reports, and packaged outputs.

ACTIVE TOOL CHAINS

Profile	Core processing	QC / interpretation
16S / ITS	QIIME 2, DADA2, SILVA classifier, MAFFT, FastTree	FastQC, MultiQC, diversity metrics, custom R/Python reports
WGS	Trimmomatic, SPAdes, Bowtie2, samtools, Prokka	QUAST, BUSCO, Kraken2, Abricate/ CARD/VFDB, eggNOG-mapper
Hybrid	NanoFilt, NanoPlot, Flye or Unicycler, BWA, Pilon, minimap2	QUAST, BUSCO, Prokka, CGView, BLASTn, Abricate, eggNOG-mapper

IMPLEMENTATION AND TECHNICAL REVIEW

- Connected shared analysis, container and server settings with QC, logs and HTML report outputs across the four execution profiles.
- Curated complex WGS/hybrid contigs using Bandage graphs, BLAST and short/long-read depth; explained assembly-graph and BUSCO evidence in technical responses.

EVIDENCE BOUNDARY

Only tools connected to active Snakemake includes and rules are listed as active. Configured or documented but inactive components are not represented as production use.

PROJECT 02 · OPERATIONS

Analysis Operations and Quality Assurance

OPERATING MODEL

Question
biological scope

→

Execution
workflow + QC

→

Delivery
interpretation + provenance

RESPONSIBILITIES AT SANIGEN

- Led NGS analysis services and team operations, combining analysis design, hands-on work and result review with SOPs, turnaround management, quality control and client technical support.
- Performed microbial-genomics, microbiome and transcriptome analyses and developed shared workflows for standardized execution, quality control, annotation and reporting.
- Developed comparative-genomics and k-mer marker-discovery tools and contributed NGS analyses to pathogen-panel research, two collaborative papers and a joint patent.
- Supported external collaborations through RNA alignment and coverage QC and evaluation of Nanopore and Ion Torrent analysis methods.
- Maintained Linux/HPC analysis environments and data-transfer infrastructure and prepared analysis manuals and staff training.

SELECTED ANALYSIS AND DELIVERY OUTPUTS

- Delivered ANI, AMR, virulence-factor and MLST comparisons, panX core-gene comparisons and SVG/Newick trees, plus featureCounts gene-count and alignment summaries.
- Corrected missing eggNOG functional-category, orthology and pathway report sections by updating the output script.
- Delivered RNA reference-alignment, target/non-target and coverage summaries for external collaboration; evaluated Nanopore Dorado modes and duplex QC.
- Prepared analysis and panX manuals and training materials while maintaining Linux/HPC environments and data-transfer infrastructure.

TRACEABILITY AND INTERPRETATION

Retained communications and analysis outputs support these examples, without a complete workload total. Gene-count delivery is separate from completed differential-expression or enrichment analysis. Platform evaluation and RNA QC do not establish regulatory qualification or therapeutic efficacy.

From Genome Differences to Assay Candidates

COMPARATIVE-GENOMICS MARKER DISCOVERY

Objective. Identify a region that distinguishes a target group while remaining practical for downstream marker or assay development.

Approach. Curate complete genomes, compare whole-genome and pan-genome context, narrow candidate intervals, and review conservation, specificity, and downstream constraints.

Outcome. Prioritized an approximately **17 kb candidate region** for follow-up, with traceable reasoning from genome selection to practical assay constraints.

FOODBORNE-PATHOGEN NGS PANEL

- Developed comparative-genomics and k-mer candidate-marker tools and reviewed panel-analysis code and QA/QC.
- Contributed NGS data analysis and interpretation to collaborative panel development and evaluation.
- Supported interpretation of NGS outputs across food and environmental sample contexts.
- Collaborative outcomes included two peer-reviewed papers in 2023 and one joint patent contribution in 2022.

BOUNDARIES

> Computational prioritization does not establish diagnostic sensitivity or specificity. Publication and patent records document collaborative outcomes, not sole authorship or universal performance across unevaluated matrices.

SELECTED EVIDENCE

- **Frontiers in Microbiology** 14 (2023): agricultural wastewater pathogen panel.
- **Journal of Microbiology and Biotechnology** 33(1) (2023): fermented-food pathogen panel.
- Joint patent contribution (2022): NGS-based primer set and foodborne-bacteria detection method.

PROJECT 05 · CURRENT ROLE CASE STUDY

Klebsiella Phage Genomics and Tail-RBP Design

GENOME EVIDENCE WORKFLOW

Host + phage genomes
QC + annotation

→

Independent evidence
taxonomy + lifestyle + termini

→

Tail interpretation
KL target + structure

CONTRIBUTION AND EVIDENCE

- Integrated CheckV, Pharokka, Phold, BACPHLIP, PhaTYP, PhaGCN, and PhageTerm outputs while preserving the evidence behind each conclusion.
- Built a tail/RBP and KL-type reference resource with source traceability, confidence labels, and experimentally supported targets explicitly flagged.
- Cross-validated candidate depolymerases with DposFinder, SpikeHunter, DepoScope, sequence search, domain boundaries, and protein-structure evidence.
- Used ProstT5 and Foldseek with experimental PDB references to identify structurally conserved proteins at sequence identities below ordinary search sensitivity.

ANONYMIZED CASE-STUDY RESULTS

- Four genomes converged on lytic **Przondovirus** classifications and short DTR packaging with 180-183 bp terminal repeats.
- Three capsule targets matched experimentally supported references at 95.4-98.8% sequence identity; unsupported cases remained unresolved rather than receiving forced labels.
- One design candidate passed genus, attachment, target-label, experimental-structure, and predicted-structure criteria; its 129-aa anchor alignment spanned 99.2% of the region.

DESIGN OUTPUT AND BOUNDARY

The design bench supports domain swaps, full-module replacements, and tandem additions, with genome FASTA, module multi-FASTA, GenBank, and protein FASTA export. Host K-locus assignments, predicted RBP folding, adsorption, host range, and lytic activity still require bacterial-genome and wet-lab validation.

PROJECT 06 · WITTGENBIO RESEARCH

CRC Atlas and Cell-Type Annotation

CRC SINGLE-CELL ATLAS WORKFLOW

Single-cell datasets
QC + metadata

→

Integrated CRC atlas
harmonization + clustering

→

Cell-type annotation
labels + validation

MULTI-STUDY DATA CURATION AND ANNOTATION

- Standardized public-study metadata, performed cell-level QC, integrated studies, and clustered cells for CRC atlas annotation.
- Cross-checked original author labels and marker evidence against tumor/normal metadata; retained source provenance and version-specific label decisions.
- Produced integrated/annotated data, marker/annotation summaries and version-comparison tables.

TUMOR AND NORMAL-TISSUE EXPRESSION EVIDENCE

CRC atlas
tumor expression

→

Target candidate review
patient / specimen evidence

←

External normal atlas
tissue expression

- Annotated externally supplied normal-tissue atlas data and reviewed tissue-specific expression.
- Combined tissue-specific expression with tumor and patient/specimen evidence to evaluate therapeutic-target candidates.
- Summarized candidate expression, coverage and interpretation limits in comparison tables with follow-up validation items; reviewed raw-count availability and depth differences.

PUBLIC EVIDENCE BOUNDARY

These analyses support RNA-level candidate prioritization. Transcript expression does not establish cell-surface protein abundance, clinical safety, or treatment response. Private datasets and internal identifiers are not disclosed.

PROJECT 07 · INDEPENDENT R&D PROTOTYPE

Phage Lifestyle Evidence Prototype

RESEARCH QUESTION

Can existing tools be combined into an end-to-end system that distinguishes lytic and lysogenic phages while preserving conflicting evidence and uncertainty?

EVIDENCE DESIGN

Quality context	CheckV completeness and contamination context
Annotation	Pharokka and Phold; integrase, repressor, excision, and lysis evidence
Classifiers	BACPHLIP and PhaTYP as independent prediction evidence
Decision layer	Agreement/confidence table with unresolved cases routed to manual review

HOST-RANGE EXTENSION

A related hypothesis-ranking layer considers LPS and outer-membrane gene variation together with phage receptor and annotation evidence. It prioritizes candidates for experiment; it does not prove adsorption, infection, or lytic activity.

PUBLICATION-GRADE NEXT STEPS

- External, leakage-controlled benchmark with known lifestyle labels.
- Comparison against individual tools and transparent simple baselines.
- Sensitivity, specificity, MCC, calibration, and quality-stratified performance.
- Ablation analysis for gene markers, classifiers, and quality context.
- Experimental validation for host-range candidate claims.

CONTACT

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